F78PA Probability & Statistics A

Tutorial 3

1. A salesman sells goods by telephone. The probability that any particular call achieves a sale is 0.08, independently of all other calls. Compute the probability that the fifth call made is the first to achieve a sale.
2. A certain basketball player in the NBA makes 70% of his free throws. What is the probability that he makes his first free throw after four attempts?
3. The number of babies, X, born during an hour at a particular maternity unit follows a Poisson distribution with mean of 4.

   i. Write down the probability mass function of X.

   ii. Compute $\mathbb{P}(X \leq 2)$.

   iii. Compute $\mathbb{P}(X \geq 4)$.

   iv. Compute $\mathbb{P}(X \geq 4 \mid X \geq 3)$.

4. A continuous random variable X has probability density function $f(x)$ given by

$$f(x) = \begin{cases} k(3+2x-x^2) & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

   for some constant k.

   i. Find the value of k.

   ii. Calculate $\mathbb{P}(1/2 \leq X \leq 3/2)$.

5. Suppose that X is a random variable with density function given by

$$f_X(x) = \begin{cases} xe^{-x} & 0 \leq x < \infty, \\ 0 & \text{otherwise.} \end{cases}$$

   (a) Sketch the density function f_X and determine $\mathrm{Im}X$, the image of X.

   (b) By integrating f_X, determine which is greater, $\mathbb{P}(X \leq 3)$ or $\mathbb{P}(X \geq 3)$.

6. Determine which of the following functions are probability density functions. Give reasons for your answers. In addition, for each density function, determine the image and the distribution function of the corresponding random variable *Hint:* Sketch the graph of each function.

   (a)

$$f(x) = \begin{cases} \frac{4}{15}x^3 & -1 \leq x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

   (b)

$$f(x) = \begin{cases} 1 & 0 \leq x \leq \frac{1}{2}, \quad 1 \leq x \leq \frac{3}{2}, \\ 0 & \text{otherwise.} \end{cases}$$

   (c)

$$f(y) = \begin{cases} y \sin y & 0 \leq y \leq \pi/2, \\ 0 & \text{otherwise.} \end{cases}$$

7. Buses arrive at fifteen minute intervals starting at noon. Anne arrives at the bus stop X minutes after noon, where X is a random variable with distribution function

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} 0 & x < 0, \\ x^2/3600 & 0 \leq x \leq 60, \\ 1 & x > 60. \end{cases}$$

(a) What is the probability that Anne waits less than five minutes for a bus?

(b) What is the probability that Anne waits more than ten minutes for a bus?

(c) Is Anne more likely to arrive before 12.30 pm or after 12.30 pm?

8. The median of a continuous random variable with distribution function F is the value m such that $F(m) = 1/2$. Find the median of X in each of the following cases.

   (a) X is exponential with parameter $\lambda > 0$

   (b) X has density given by $f_X(x) = 2xe^{-x^2}$ for $x > 0$ (and $f_X(x) = 0$ otherwise).

9. Let X be a random variable with density

$$f_X(x) = \begin{cases} \frac{x^2}{9} & 0 \leq x \leq 3, \\ 0 & \text{otherwise.} \end{cases}$$

   (a) Find $\mathbb{E}[X]$.

   (b) Let $Y = 1/X$. Determine $\mathbb{E}[Y]$.

   Remark: This example shows that, in general, $\mathbb{E}\left[\frac{1}{X}\right] \neq \frac{1}{\mathbb{E}[X]}$.

Exam Style Question

The lifetime of a certain brand of battery (measured in tens of hours) is modelled by the continuous random variable X with the probability density function given by

$$f_X(x) = \begin{cases} \frac{2}{75}x, & 0 \leq x \leq 5 \\ \frac{2}{15}, & 5 < x \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Explain why f_X is a density function for a random variable X. Give reasons for your answers. [1 mark]

(b) Determine the image, $\text{Im}(X)$, of the random variable X and find the cumulative distribution function, F_X, of X. Give reasons for your answers. [3 marks]

(c) Hence, calculate:

   i) $\mathbb{P}(X > 4)$ [1 mark]

   ii) the median, m of the random variable X. Recall the median of a continuous random variable is the point m such that $\mathbb{P}(X \leq m) = \mathbb{P}(X \geq m)$. [1 mark]

   Assuming the lifetimes of the batteries are independent of each other. Two of these batteries are required for an electronic device to operate, and it will only if **both** batteries are operational.

(d) If two batteries are installed in this device, determine the probability that the device will stop working within the next 40 hours. [2 marks]

[8 marks]